

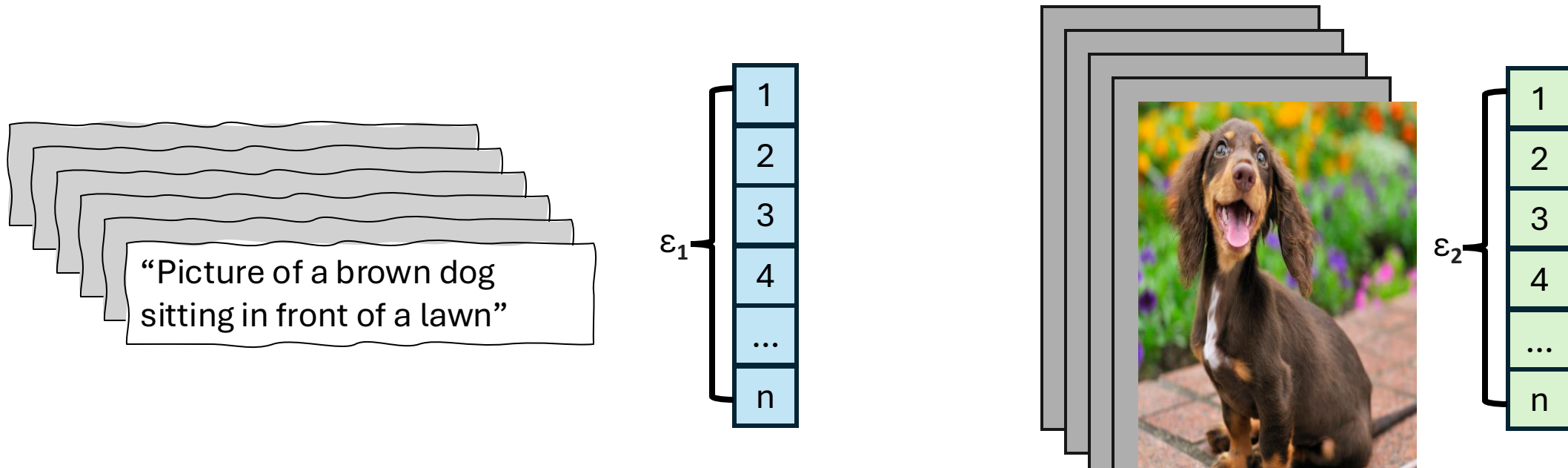
Image Classification: Contrastive Language Image Pretraining

SuperCLIP
By Wu and Caputa

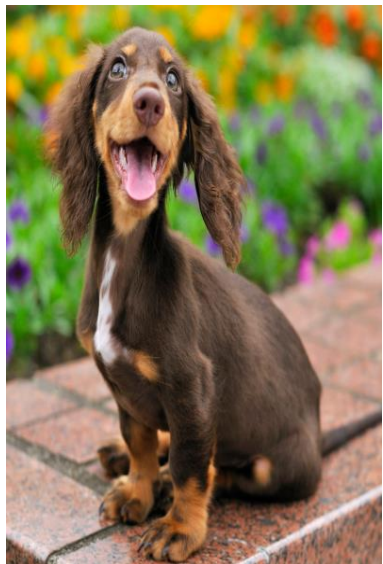
WHAT IS CLIP?

The CLIP method trains 2 models contrastively:

- One model takes in a piece of text as input and outputs a single vector representing its semantic content. (text)
- Another model takes in an image and similarly outputs a single vector representing its visual content. (image)



Embedded Space

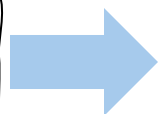


ε_2

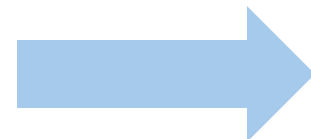


V

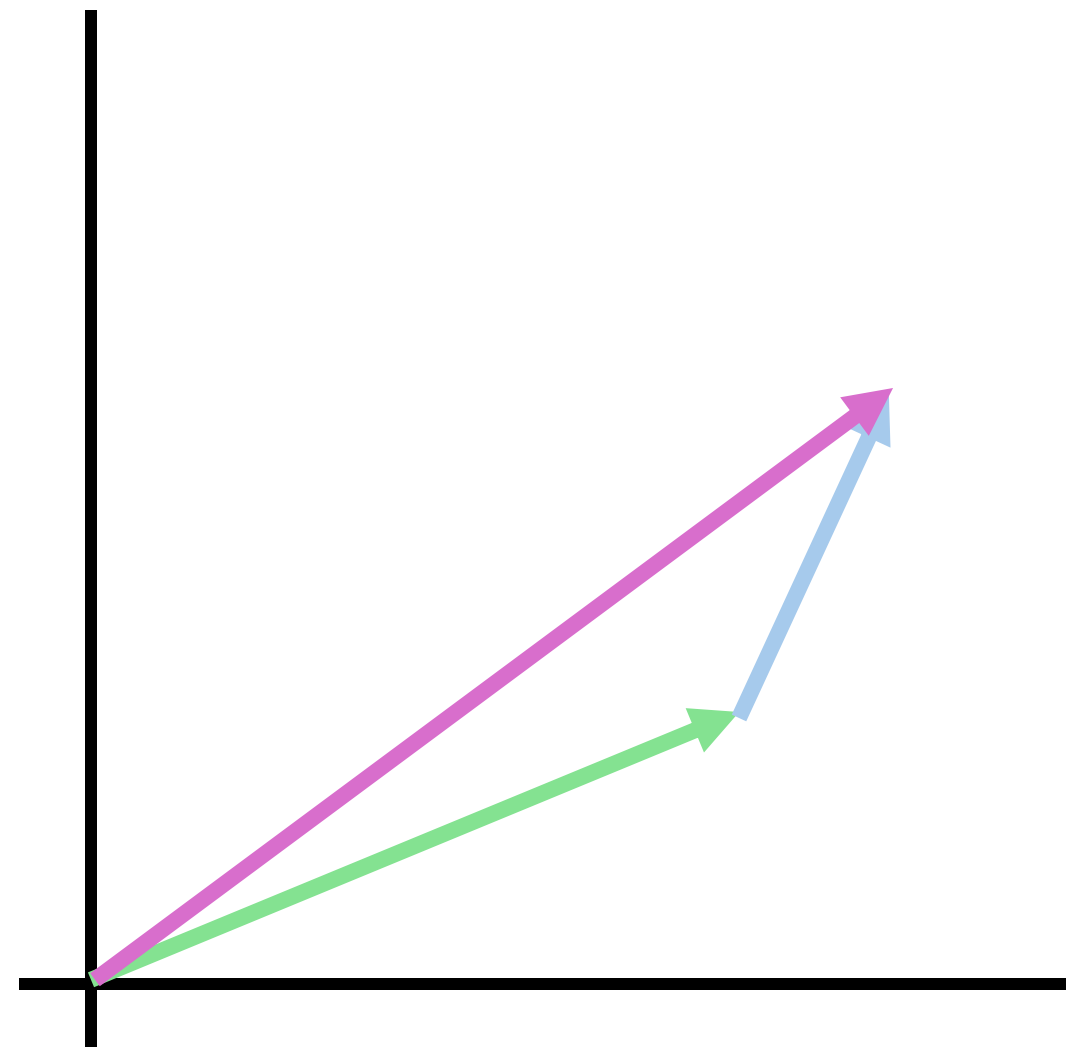
“Picture of a brown dog
sitting in front of a lawn”



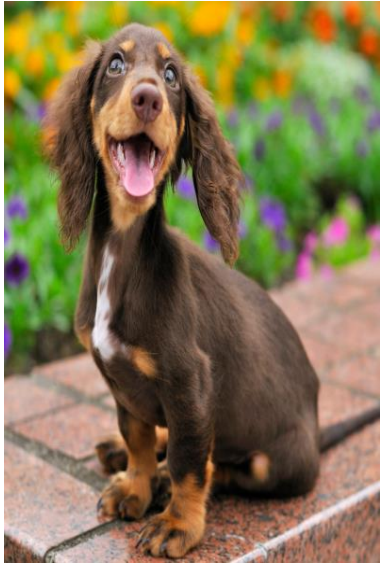
ε_1



V



Embedded Space

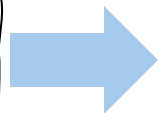


ε_2



V

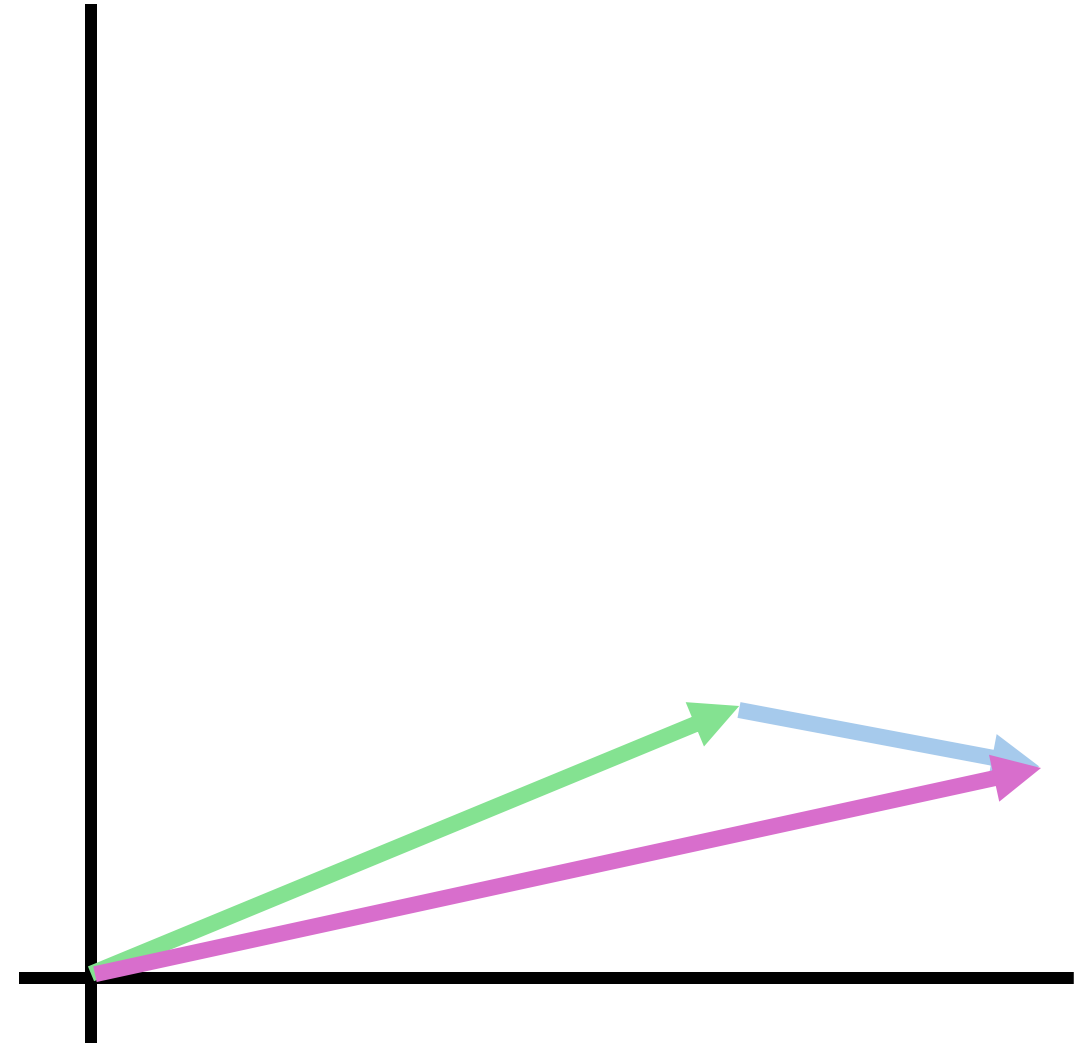
“Picture of a brown cat sitting in front of a lawn”



ε_1



V



Embedded Space



“Picture of a brown dog sitting in front of a lawn”						
ϵ	1	2	3	4	...	N
1	d1					
2		d2				
3			d3			
4				d4		
...					...	
N						d5

By comparing every image embedding with every text embedding in a batch using cosine similarity, CLIP computes the distance between each pair creates a similarity matrix, where pairs lie on the diagonal.

Zero Shot

New Image



ε_2



“Picture of a brown dog
sitting in front of a lawn”

ε	1	2	3	4	...	N
	d1					
		d2				
			d3			
				d4		
					...	
						d5

At inference, CLIP compares a new image to text descriptions and selects the most similar one.

Drawbacks

CLIP is powerful, but it relies heavily on text prompts, lacks strong class separation, and compresses images, losing fine details

Prompt sensitivity

Weak class structure

Loss of fine details

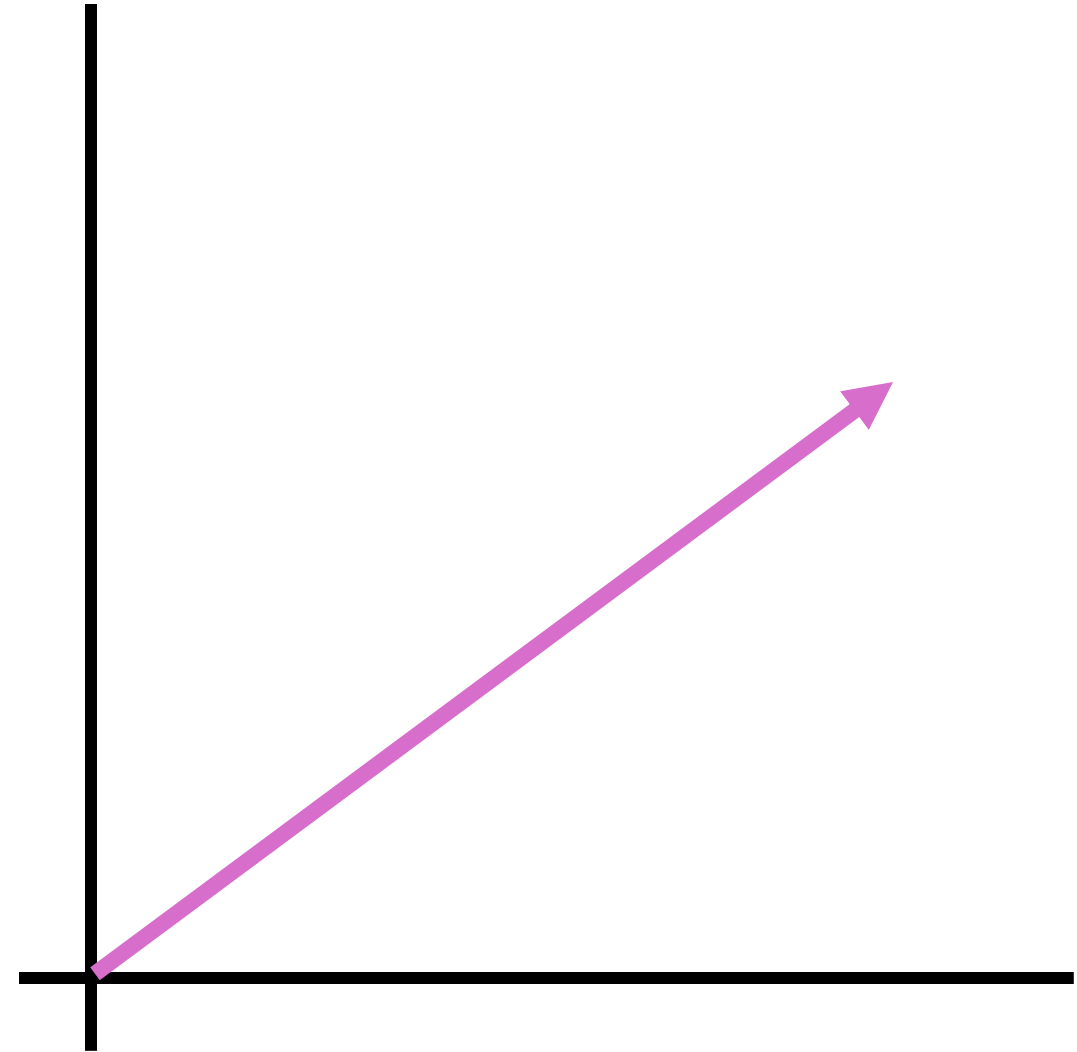
No reasoning

SuperCLIP

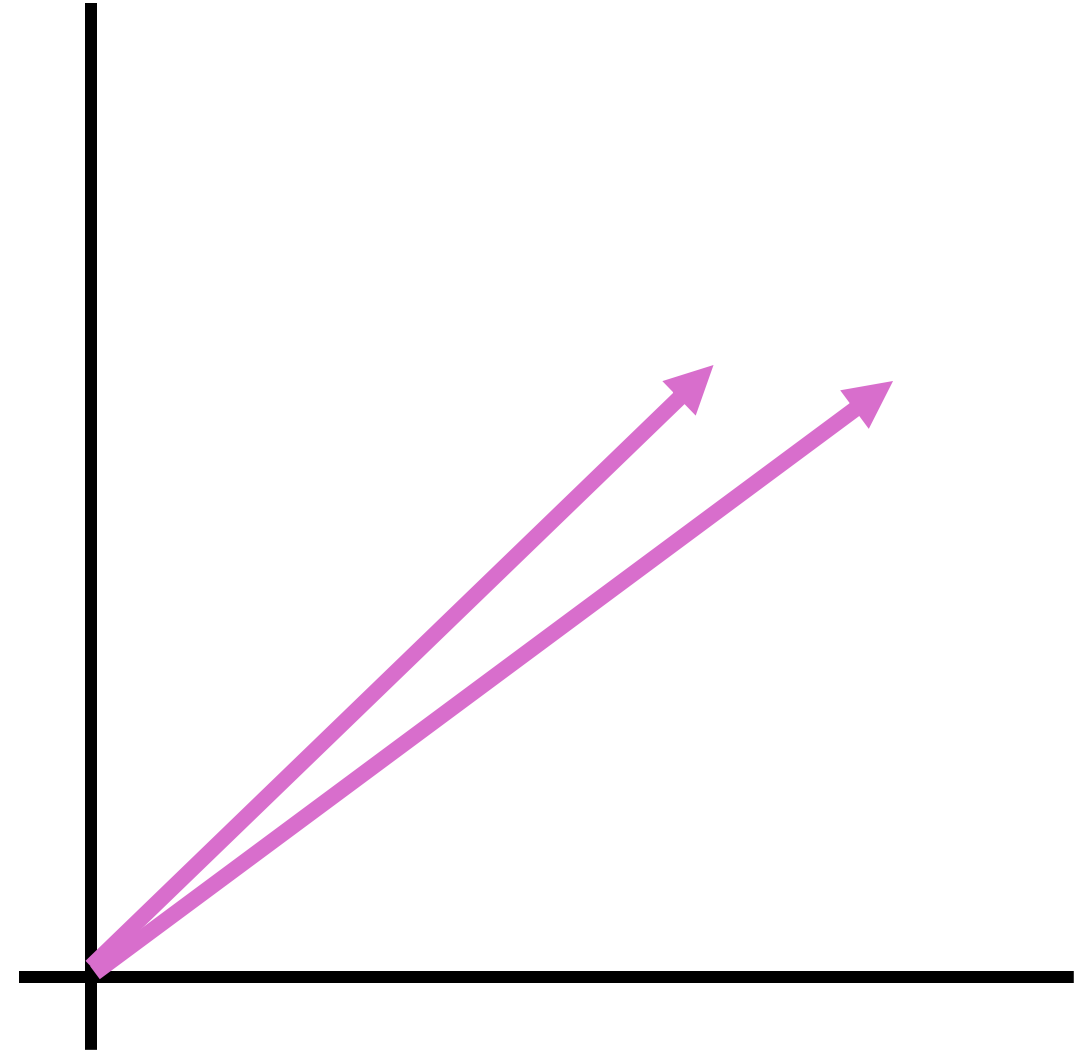
SuperCLIP keeps the contrastive objective but adds **CLASS LABELS** to organize embeddings into clearer clusters.

- Better class separation
- Reduced prompt sensitivity
- More stable embeddings

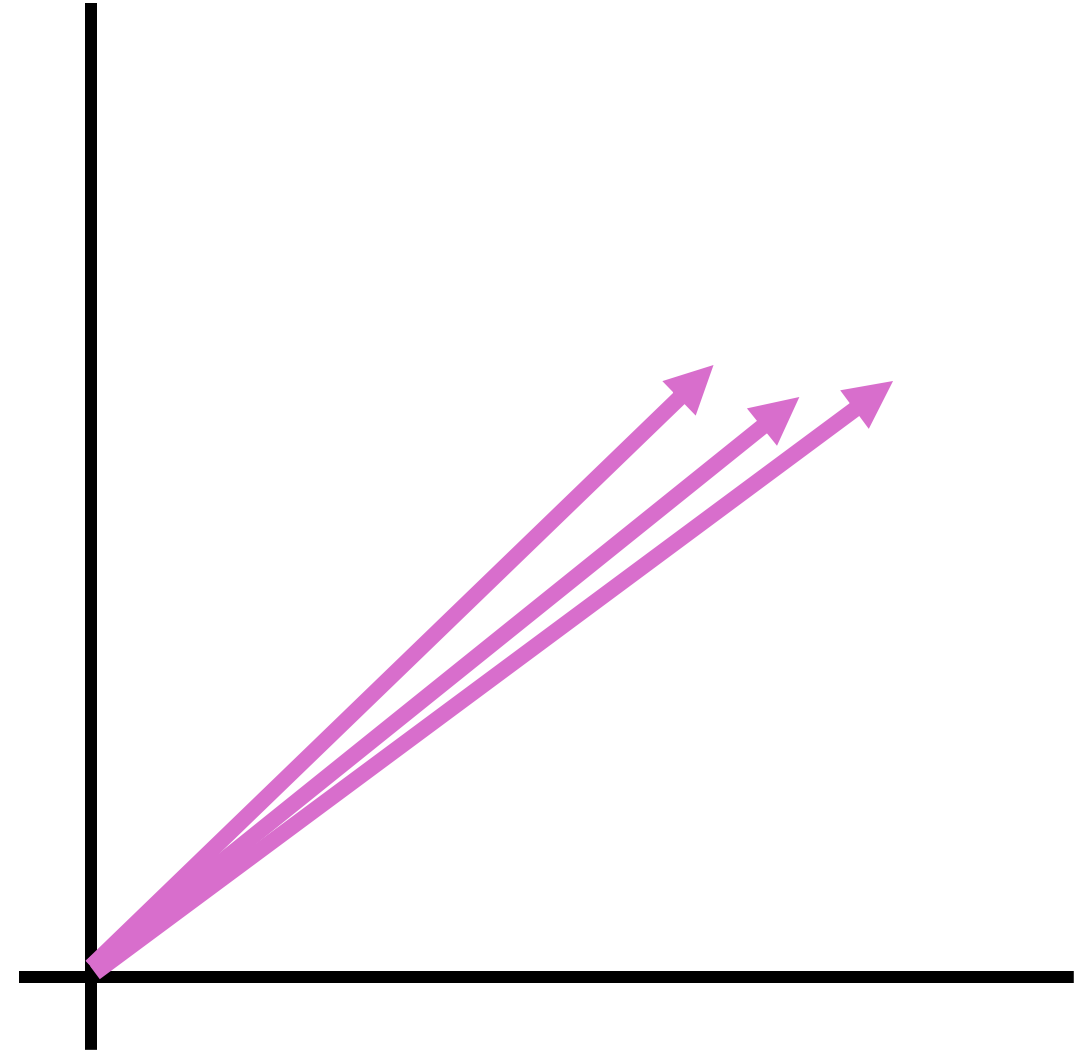
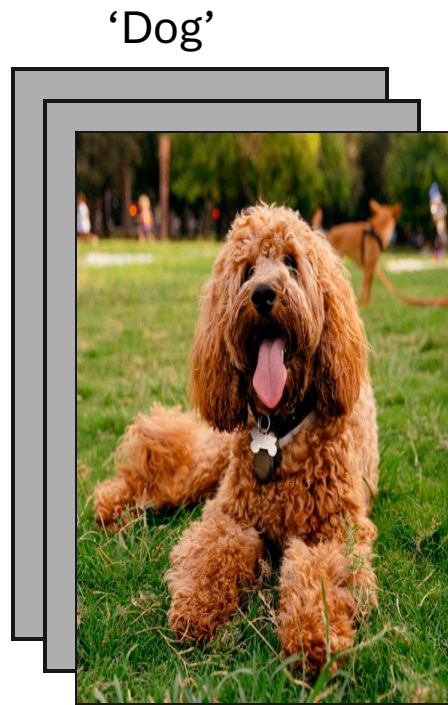
Classification



Classification

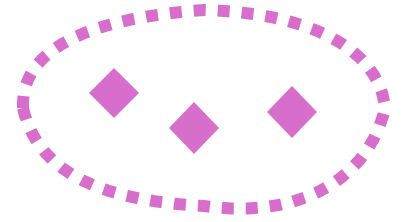


Classification

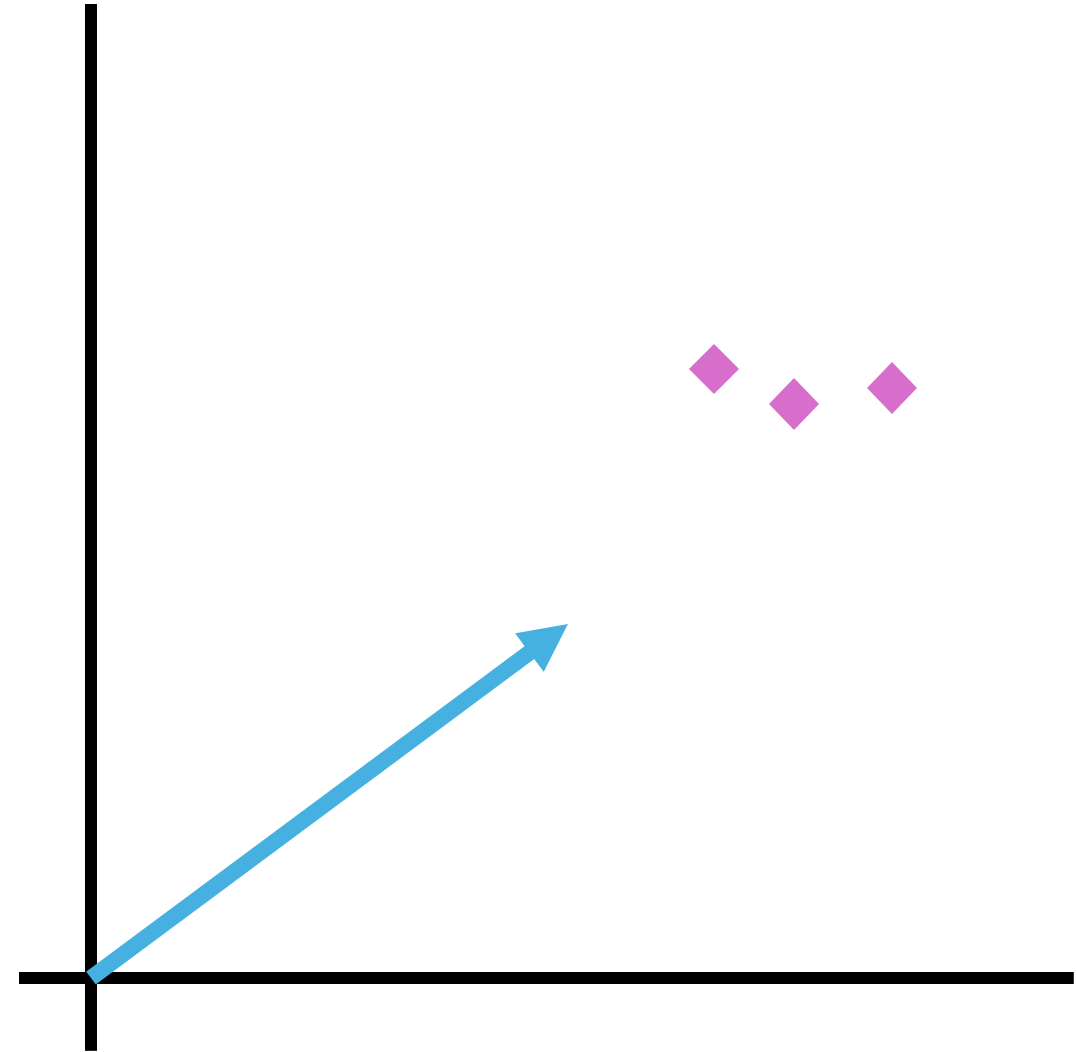
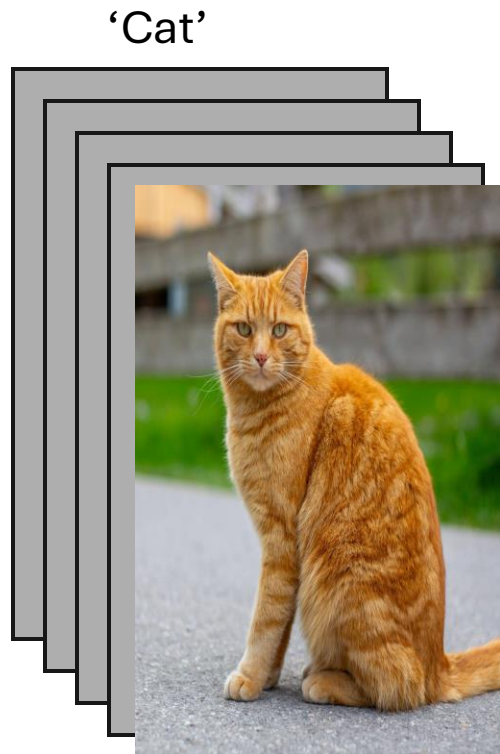


Classification

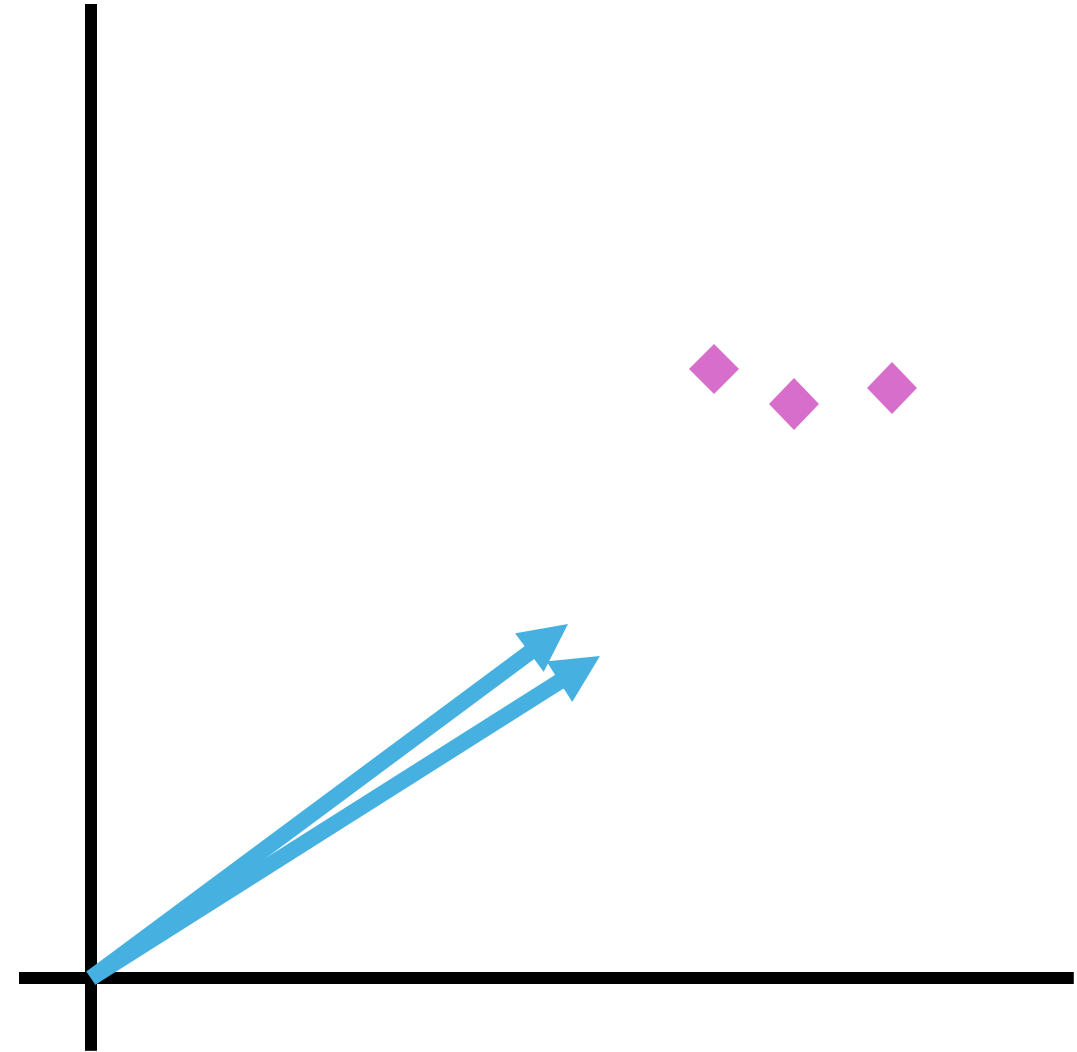
'Dog'



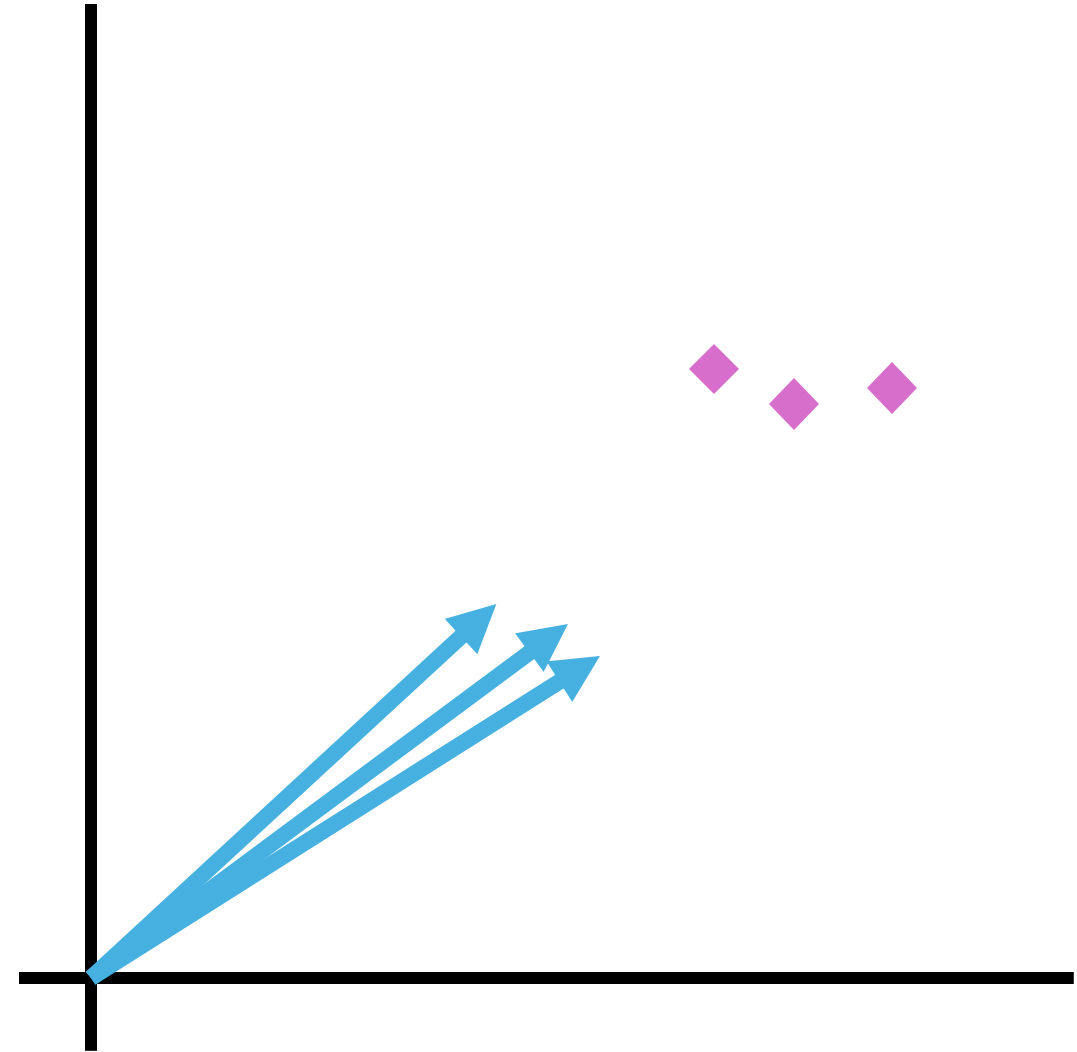
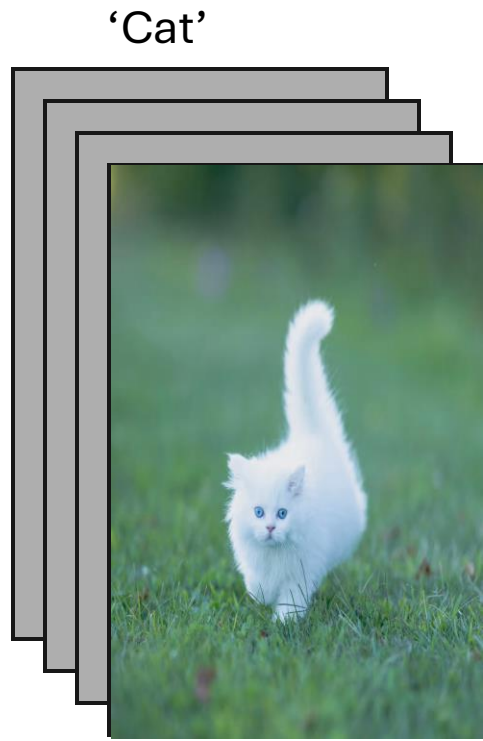
Classification



Classification



Classification

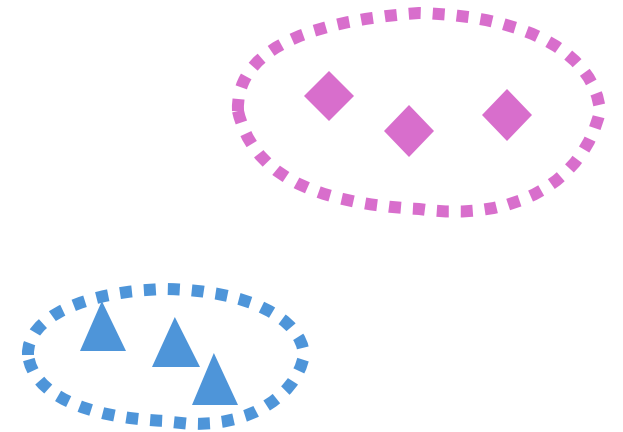


Classification

'Cat'



Classification organizes the embedding space so that samples from the same class are grouped together.



The Bag-of-Words Gap

These two captions produce nearly identical supervision in SuperCLIP:

*"A dog chasing a red ball
across a field"*

VS

*"A red ball near a dog
sitting in a field"*

Both → same pseudo-labels: { dog, chase/sit, red, ball, field }

Why this matters:

- Token classification = bag of words: no relational, attributional, or action structure
- Research (Winoground, ARO) shows CLIP-style models fail at compositional distinctions
- Retrieval and VQA suffer when representations lack phrase-level semantic structure

Our Proposal: Caption Reconstruction Auxiliary Loss

Core idea:

Add a small reconstruction head on top of SuperCLIP that forces the image encoder to predict masked or withheld caption tokens from its own visual features.

$$\mathbf{L}_{\text{total}} = \mathbf{L}_{\text{token-cls}} + \lambda \cdot \mathbf{L}_{\text{recon}} \quad \lambda \in [0.1, 0.5]$$

Variant A — Masked Token Prediction

Randomly mask 15–30% of caption tokens. Train image features to predict the masked tokens via cross-entropy over vocabulary. Inspired by BERT masking.

Variant B — Phrase Reconstruction

Extract noun phrases (e.g. 'red ball', 'chasing dog') using spaCy. Predict them from image features. Directly targets compositional structure.

The reconstruction head is a minimal 2-layer MLP — no new backbone, no extra data, negligible compute overhead.

Dataset & Training Setup

MS-COCO Captions

~118K training images

5 captions per image

5K test images (standard split)

Why COCO: rich human captions, standard retrieval splits, manageable at course scale.



Model & Training Config

Backbone: ViT-B/32 (initialized from CLIP)

Recon Head: 2-layer MLP projection

λ sweep: 0.1 $\rightarrow$ 0.3 $\rightarrow$ 0.5

Mask rate: 15–30% of caption tokens

Optimizer: AdamW with warmup LR schedule

Batch size: 128 (gradient accumulation)

Evaluation Plan

PRIMARY METRIC

Image-Text Retrieval on COCO — Recall@1, R@5, R@10 (image→text and text→image)

Baseline vs. Ours

Side-by-side retrieval. Directly answers: does reconstruction supervision help?

Compositional Probe

Winoground / ARO subset. Tests relational structure preservation.

Ablation Study

Sweep λ , masking rate, Variant A vs B. Isolates the source of gains.

Illustrative results table (to be filled after experiments):

Model Variant	R@1 (i→t)	R@5 (i→t)	R@1 (t→i)	Winoground
Baseline SuperCLIP	XX.X	XX.X	XX.X	XX.X
+ Masked Recon ($\lambda=0.1$)	XX.X	XX.X	XX.X	XX.X
+ Masked Recon ($\lambda=0.3$)	XX.X	XX.X	XX.X	XX.X
+ Phrase Recon ($\lambda=0.3$)	XX.X	XX.X	XX.X	XX.X

Summary

01

Baseline

Reproduce SuperCLIP token-classification pipeline on MS-COCO as the anchor.

02

Gap

Token presence supervision is bag-of-words: it misses compositional and relational semantics.

03

Contribution

Add a lightweight caption reconstruction auxiliary loss — masked token or phrase prediction from image features.

04

Evaluation

Test on retrieval (R@1/R@5) and compositional probes (Winoground/ARO) with ablations on λ and variant.

Why it matters:

Compositional understanding is a known weakness of CLIP-style models with real-world implications in retrieval and VQA. This project tests a cheap, principled fix at course scale.